6- Software Basics :-

Data Representation :- (learn about how computer represents numbers and colors)

Are you interested how computers represent colors and numbers? you are not interested in seeing purple color on screen, but also interested in inspecting how "purple" is represented in computer memory. As bonus you will also learn how computer manage to do their all calculations when they can read and write only two states and interpret them as 0 and 1.

1- light on 0 - light off

Human beings find it most convenient to use decimal system for everyday tasks, such as counting, measuring, weighing, and paying. If you like jogging, you might track your progress by timing yourself; for instance you might jogged for 23 minutes. when you look at prices, you expect something like 17, 99, 239, 3049 and so on. without giving it much thought we use decimal systems, where digit range from 0 to 9. Computers on other hand they are limited to 2 digit 0 and 1. If you have ever wondered how computers represent different values using only 2 digits this room will get you answer.

Learning Objectives :-

* Representing 8 colors * Representing 16 million colors * Binary numbers * Hexadecimal numbers
* (optional) Octal numbers

Task 2 :- Representing Colors

Our first Eight colors :-

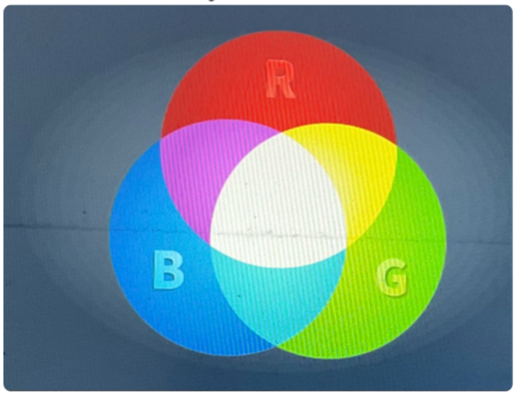
By combining different amounts of red, green, blue lights, you get any color you like. In computer colors think of it like 3 knobs, each knob controls one of the 3 colors.

Example :-

let's say that each of 3 colors can be either on or off i.e each has 2 states.

* The red light can be either on or off
* The green light can be either on or off
* The blue light can be either on or off

These states gives us 2×2×2 = 8, thats eight different colors



If computer were limited to 8 colors, It would only need to Indicate which color is "Switched on" and which is "Switched off". It can use 3 digits of 1 and 0 to represent states of red, green and blue. For Example, 111 would be all 3 lights switched on, while 100 would be only red switched on. The digit which can be either 1 or 0 is called a bit.

A computer can represent any of 8 colors. If its still unclear detailed list is shown in below.

Color Representation	Representation Meaning	Color Name
000	All colors are off	Black
001	only blue color is on	Blue
010	only green color is on	Green
100	Only red is on	Red
011	Green and Blue are on	Cyan
101	Red and Blue are on	Magenta
110	Red and green are on	yellow
111	All colors are on	white

we just provided a way to represent color in palette of 8 colors.

From 8 to 16,000,000

Being limited to 8 color is inconvenient, As prefer millions of colors. If each 3 lights (red, green and blue) had 256 levels Instead of just 2 (on or off). $256 \times 256 \times 256 = 16,777,216$. That's more than 16 million colors.

* one bit is enough to represent 2 states: on and off. we need 8 bits to express 256 states.

In most textbooks a group of 8 bits is referred as byte; however you can also use term octet.

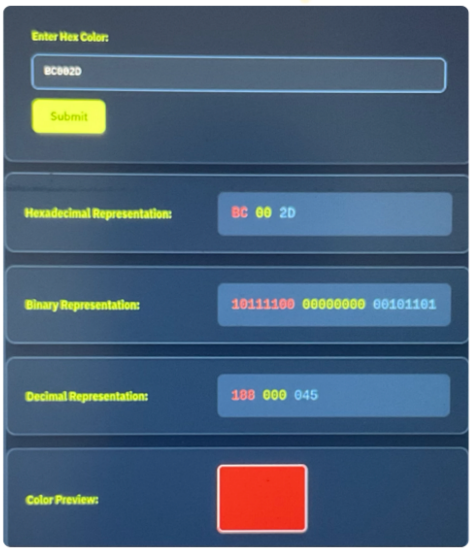
& 3×8 bits or 3bytes (24 bits). Ex:- One green color used on this page is represented as 101000111110 1010 00 101010; that's not convenient way to read color codes. Here comes Hexadecimal to rescue

Hexadecimal Representation:-

Hexadecimal representation makes it easy to combine 4bits into single character, A Hexadecimal digit, to be specific. For now think Hexadecimal digits as symbols where each symbol represents set of 4bits.

Hexadecimal digit	Binary Representation
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Going back to Green color, Instead of typing 10100011 11101010 00101010, we can type A3EA2A.



To Summarise:-

* In real life applications, a color is represented in 24 bits i.e 3bytes.
* Each byte can represent 256 different values.
* Each of 3bytes specifies intensity of red, green and blue lights.
* Every 4bits are represented by one hexadecimal digit
* Each byte is represented as 2 hexadecimal digits.

Quiz:- view site and use this tool

1) Preview the color #3BC81E. what color appears to be?

A) Green

2) what is binary representation of color #EB0037?

A) 11101011 00000000 00110111

3) what is Decimal representation of color #D4D8DF?

A) 212 216 223